

# Technology Stack

|              |                                                    |
|--------------|----------------------------------------------------|
| Team ID      | SWUID2026383049                                    |
| Project Name | EduGenie: Google Gemini Powered Learning Assistant |
| Team Size    | 4                                                  |
| Team Leader  | Ezhumalai V                                        |
| Team Member  | M Monishkumar                                      |
| Team Member  | Boopathi Pandiyan P                                |
| Team Member  | Nithish D                                          |
| Date         | 30 September 2026                                  |

## Technology Stack (Architecture and Stack)

| S.No | Component           | Description                                  | Technology                                                                   |
|------|---------------------|----------------------------------------------|------------------------------------------------------------------------------|
| 1    | User Interface      | Responsive web UI with forms and quiz runner | HTML5, CSS3, JavaScript, Jinja2                                              |
| 2    | Application Logic-1 | REST API and routing                         | Python, FastAPI, Uvicorn                                                     |
| 3    | Application Logic-2 | Q&A, Summary, Quiz, Learning Path modules    | Python modules (qna.py, summary_module.py, quiz_module.py, learning_path.py) |
| 4    | Application Logic-3 | Concept explanation with fallback            | LaMini-Flan-T5-783M (Transformers, PyTorch) and Gemini                       |
| 5    | Generative AI Model | Text generation                              | Google Gemini 1.5 Pro / Flash                                                |
| 6    | Configuration       | API key and settings                         | python-dotenv (.env)                                                         |
| 7    | Data Storage        | No database, stateless requests              | Not applicable                                                               |
| 8    | Infrastructure      | Local / cloud deployment                     | Uvicorn (local), any cloud VM or PaaS                                        |

## Application Characteristics

| S.No | Characteristic           | Description                                       | Technology                             |
|------|--------------------------|---------------------------------------------------|----------------------------------------|
| 1    | Open-Source Frameworks   | FastAPI, Uvicorn, Jinja2, Transformers            | Python ecosystem                       |
| 2    | Security Implementations | API key kept in environment file; CORS configured | python-dotenv, FastAPI CORS middleware |
| 3    | Scalable Architecture    | Modular, stateless services                       | FastAPI modules                        |
| 4    | Availability             | Model fallback for resilience                     | Gemini Pro to Flash to local T5        |
| 5    | Performance              | Lightweight prompts and concise outputs           | Gemini Flash / Pro                     |